

Synthetic data boosts medical foundation models

Bin Sheng, Pearse A. Keane, Yih-Chung Tham & Tien Yin Wong



Using synthetic data generated via conditioning with disease labels can enhance the pretraining efficiency and generalization of medical foundation models, as shown for the detection of eye diseases via fundus photographs.

Medical foundation models are built to generalize across medical domains and tasks^{1,2}. Yet the performance and generalization of these foundation models can be hindered by challenges in the acquisition of vast medical data. A major difficulty is the often-insufficient amount of data that can be obtained through open-source datasets. Another difficulty is that large and well-annotated datasets are typically safeguarded within health systems and are therefore not openly accessible. Concerns regarding patient privacy, anonymization and the high costs associated with the collection of clinical data further compounded these challenges. One solution to these challenges is to build foundation models without direct data sharing by using federated learning. Writing in *Nature Biomedical Engineering*, Bo Yan and colleagues now describe an easier approach: leveraging the generation of synthetic

medical images tailored to specific disease labels³. At the heart of the study by Yan and colleagues lies the creation, via diffusion models, of a synthetic dataset comprising more than one million colour fundus photography images. By fine-tuning a pretrained stable diffusion model using a limited set of real-world fundus images annotated with disease labels, the researchers generated synthetic retinal images that mirror the intricacies of their real counterparts.

Specifically, Yan and colleagues pretrained a foundation model using both the synthetic dataset and a smaller subset of real-world images (Fig. 1). The two-stage pretraining process enhanced the model's ability to extract meaningful features from unlabelled data, as evidenced by its performance on tasks ranging from diabetic retinopathy grading to multi-disease classification. Notably, the model's performance was comparable or exceeded that of models pretrained exclusively on massive real-world datasets. This result underscores that synthetic data can be a valuable supplement to limited real-world data and suggests that medical artificial intelligence (AI) models may also perform well in data-scarce environments.

A key advantage of the approach by Yan and colleagues lies in its ability to improve efficiency in data curation for pretraining purposes. By leveraging synthetic data, the model required substantially fewer (~10%) real-world images during pretraining while maintaining or surpassing the performance of a foundation model for generalizable

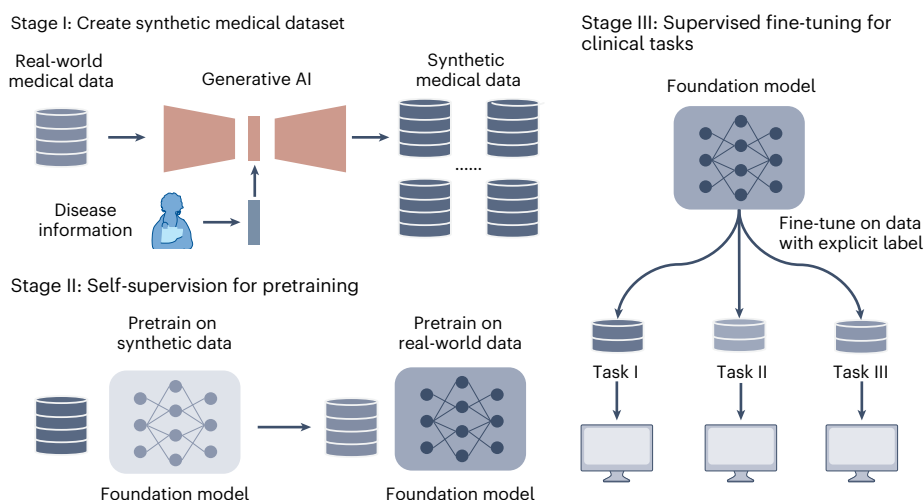


Fig. 1 | A data-efficient framework for developing medical foundation models through synthetic-data generation, self-supervised pretraining and supervised fine-tuning. In stage one, real-world medical data and disease-specific information are used to train a generative AI model that synthesizes large-scale medical datasets. These synthetic datasets serve as a diverse and scalable resource for pretraining. In stage two, a foundation model is first pretrained using synthetic data, enabling it to learn generalizable medical

representations. This is followed by additional pretraining on real-world medical data, which refines the model's understanding of clinically relevant features. In stage three, the pretrained foundation model is adapted to specific clinical tasks through fine-tuning on labelled datasets. This step tailors the model for real-world applications such as disease classification, diagnosis support and medical image interpretation. Figure adapted from ref. 3, Springer Nature Limited.

disease detection from retinal images (RETFound²) that was pretrained solely via real-world images. Furthermore, the authors' model showed remarkable label efficiency – that is, the percentage of labelled training data needed for the model to achieve a target performance – thus minimizing the reliance on expert annotations. These findings suggest that synthetic data can alleviate the burden of annotating vast image collections, which would accelerate the development and deployment of medical AI models.

The study by Yan and colleagues also describes their model's enhanced generalization capabilities. When evaluated on datasets from different geographical locations and imaging devices, the model consistently outperformed baseline models, which indicates its resilience to domain shifts and hence the potential of synthetic data to mitigate biases inherent in real-world datasets and to facilitate the development of foundation models that can be possibly applied broadly across diverse medical settings. Moreover, the ability to create balanced synthetic datasets may address long-standing issues related to data imbalance and to promoting fairness and inclusivity in medical AI.

The work by Yan and colleagues raises a few important questions. First, although AI-generated data may reduce the privacy risks associated with real-world medical data, it does not remove them entirely. Second, the black-box nature of foundation models obscures the reasons for performance deterioration or failure when the models are trained primarily or solely on synthetic data; this shortcoming can leave developers and users uncertain about the extent of any 'toxicity' from AI-generated data. Third, guiding synthetic-data generation with a limited sample of real-world disease labels may unintentionally reinforce biases inherent in small real-world datasets, thereby compromising the model's equity, fairness and generalizability, especially for rare diseases or for ethnic minorities and other under-represented groups. It remains unclear how much real-world data are necessary to construct a robust foundation model; in fact, the performance of a model trained entirely on synthetic data remains unknown. Moreover, it is imperative to establish guidelines and standards that ensure the traceability and provenance of both real-world and synthetic data in medical AI.

Yan and colleagues' work also raises fundamental questions about the role of AI-generated data use in building foundation models. Most importantly, AI-generated data can act as a catalyst or as a contaminant of the performance of medical foundation models. By exploring the feasibility of leveraging synthetic data to address data scarcity and improve model generalization, the authors have explored a path forward for potentially more efficient medical AI research. It is thus

imperative to navigate the delicate balance between the opportunities offered by synthetic data and the need for rigorous validation, ethical considerations and a commitment to continuous improvement in the collection of real-world data.

Moreover, current AI models are far from reaching the status of a 'universal model' that captures the complexities of human health^{4,5}. Establishing foundation models for universal health problems necessitates a holistic approach that transcends sole reliance on AI-generated data. Given the vast unknowns in human biology and health, enhancing the efficiency and universality of real-world data collection remains paramount. Synthetic data, when used judiciously alongside real-world data, can aid in this endeavour but are unlikely to serve as a standalone solution.

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Competing interests

The authors declare no competing interests.